

Rajiv Harlalka

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Education

Indian Institute of Technology Kharagpur

Integrated M.Sc. Mathematics and Computing (CGPA: 8.94/10)

Kharagpur, India

2020 – 2025

Work Experience

Meta, London | Software Engineer

Sept 2025 - Present

- Designed and implemented an Agentic UX compliance flow to regularly validate **over 500 active user consents**, ensuring strict design compliance **eliminating 40 work-week audit annually**
- Implemented **German government-mandated account** discovery settings and timeline visibility setting across all Meta platforms, ensuring regulatory compliance for all new German user accounts
- Integrated automated user consent checks into the data lineage pipeline across **1,000+ sensitive data assets**, to audit global data retention and processing preferences, **reducing audit costs by 88%**

Microsoft, India | Software Engineer Intern

May 2024 - Jul 2024

- Worked on enabling **edits on high-fidelity fields** for tasks on “Assigned to Me” View of **Microsoft Planner**.
- Introduced a dynamic user permission model for API enhancement and verified data convergence between sources.

Houseware | Backend Intern

Aug 2023 - Oct 2023

- Implemented organization management endpoints on the app for administrators to manage access to active users.
- Revamped **DBT models** to handle incremental changes for **Snowflake & BigQuery warehouse**, thus reducing compute.

Mello | Backend Intern

Feb 2022 - Dec 2022

- Designed and built the principal backend and its infrastructure that served **1000+ users** on the mobile application.
- Implemented **write-through cache** using Redis & asynchronous request handling to **reduce response time by 20%**.
- Introduced internal tools for improved user communication, admin-side registration tracking and on-site user verification.

Projects

AlgoStat: Live Algo-Trading Platform for NSE Derivatives

June 2025 – Present

- Built tick-to-trade pipeline for Indian derivatives market ingesting **~500 live tickers**, with **~20ms e2e latency**
- Built Bloomberg style data plane to publish and compare public and custom developed indices on a daily cadence
- Developed **vectorized backtester** for options/futures/equity baskets using daily and custom calendars; runs **10 years of history in <60s** for strategy validation

FileDB: A Toy Disk Based Key-Value Storage

March 2025 - June 2025

- Developed a disk persistent key-value store inspired by Bitcask Paper from Riak with order **O(1) record reads**.
- Implemented background file compaction similar to LSM-trees, to reclaim space and sync write buffers to disk
- Provides high write throughput due to append-only writes and reached a benchmark of **50k write/s and 110k read/s**

Google Summer of Code 2024 — FFmpeg

May 2024 - Aug 2024

- Engineered a video filter in FFmpeg to support **PU21 encoding** for HDR image and video quality assessment.
- Engaged with the community to understand the fundamentals of multimedia encoding/decoding and processing.

Competitions

OpenSoft - MongoDB Movie Website Hackathon

March 2024

- Led a team of 8 members to win the **first** position for Movie Streaming Website Hackathon by **MongoDB** for OpenSoft.
- Implemented **adaptive bitrate streaming via HLS protocol** served through CDN for minimal video buffering.
- Enhanced search functionality by incorporating **autocomplete suggestions**, fuzzy matching and semantic search.

Inter IIT Tech Meet 11.0

Jan 2023

- Member of **Gold** winning team for Path Planning for Last Mile Delivery by Grow Simplee during InterIIT Tech Meet.
- Developed **REST & gRPC services** to serve the frontend and optimization service with a P95 latency of **120ms**.
- Implemented coordinate encoding using the **Geohash** feature by Redis reducing distance computation time by **40%**.

Technical Skills

Programming Languages	C/C++, Hack, Golang, Javascript, Typescript, SQL, Bash, Perl
Frameworks & Databases	Node.js, React.js, PostgreSQL, SQLite, MongoDB, Redis
Tools & Platforms	Linux, Git/GitHub, Docker, Nginx, AWS EC2, gRPC, RabbitMQ

Positions of Responsibility

IQPS - Metakgp | Maintainer

- Revamped IQPS - Intelligent Question Paper Search, a **question paper discovery service** hosting 600+ documents.
- Implemented **vector search** using pgvector on PostgreSQL that searches through course code, year and course name.
- Developed a simple form based platform for crowd-sourcing question papers and an admin panel for detecting outliers.